EXPERIMENT 6

Automating Business Processes Using Deluge Scripting and Workflows, with Testing and Debugging

Gym Membership Management System – member welcome-mail automation

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Application	Gym_Membership_Management_System (Zoho Creator)	**Department**	B.Tech CSBS

1. Aim

To automate a key business process of the gym application, the follow-up of a member registration, using a Zoho Creator form workflow and a Deluge script. When a member is registered the script derives the membership end date, defaults the status to Active and sends a welcome e-mail with the plan details, and the automation is tested and checked through the delivered e-mail.

This removes the need for front-desk staff to calculate dates by hand and to write to each new member, so that every registration is confirmed immediately and in the same way.

2. Procedure

Step 1: Start a new workflow. From Workflow → Form workflows a workflow was created on the Member_Registration form.

Step 2: Define the trigger. The Record Event was set to Created and the Form Event to Successful form submission, so the automation fires only after a valid, saved registration.

Step 3: Add the action. A Deluge script action was added with Run this action set to Always, because the script needs conditions and a record update that a simple notification cannot provide.

Step 4: Write the Deluge script. The script has four steps: (1) fetch the selected plan by ID and derive its duration from the plan name (Basic 1, Standard 3, Gold 6, Platinum 9, Diamond 12 months); (2) calculate the end date with addMonth() only when it is blank; (3) set Membership Status to Active when it is blank; (4) send the welcome e-mail when an address was entered.

Step 5: Save and review. The script was saved and the workflow summary showed the single Deluge action under Member_Registration › Created › Successful form submission.

Step 6: Test. A member was registered with the Platinum plan and a valid e-mail address, and the inbox was checked for the welcome e-mail.

3. Output

3.1 Workflow configuration

[Screenshot to be inserted]
Insert the workflow screenshots: new workflow (form, record event, form event, name) and the Deluge script action summary.

3.2 Deluge script (Code 1)

```
// 1. Get plan details from the Membership_Plans lookup
planName = "";
planFee = "";
duration = 0;
planRec = Membership_Plans[ID == input.Membership_Plans];
if(planRec.count() > 0)
{
    planName = planRec.Plan_Name;
    planFee = planRec.Plan_Fee;
    // Duration worked out from the plan name
    pn = planName.toString().toLowerCase();
    if(pn.contains("basic"))
    {
        duration = 1;
    }
    else if(pn.contains("standard"))
    {
        duration = 3;
    }
    else if(pn.contains("gold"))
    {
        duration = 6;
    }
    else if(pn.contains("platinum"))
    {
        duration = 9;
    }
    else if(pn.contains("diamond"))
    {
        duration = 12;
    }
}
// 2. Calculate end date if it was left blank
startDate = input.Membership_Start_Date;
endDate = input.Membership_End_Date;
memRec = Member_Registration[ID == input.ID];
if(endDate == null && startDate != null && duration > 0)
{
    endDate = startDate.addMonth(duration);
    memRec.Membership_End_Date = endDate;
}
// 3. Set status to Active if it was left blank
if(input.Membership_Status == null || input.Membership_Status == "")
{
    memRec.Membership_Status = "Active";
}
// 4. Send the welcome email
if(input.Email_Address != null && input.Email_Address != "")
{
    startText = "";
    endText = "";
    if(startDate != null)
    {
        startText = startDate.toString("dd-MMM-yyyy");
    }
    if(endDate != null)
    {
        endText = endDate.toString("dd-MMM-yyyy");
    }
    sendmail
    [
        from: zoho.adminuserid
        to: input.Email_Address
        subject: "Welcome to our Gym, " + input.Member_Name + "!"
        message: "Dear " + input.Member_Name + ",<br><br>"
               + "Thank you for registering with our gym. Your membership details are below:<br><br>"
               + "<b>Membership Plan:</b> " + planName + "<br>"
               + "<b>Plan Fee:</b> Rs. " + planFee + "<br>"
               + "<b>Start Date:</b> " + startText + "<br>"
               + "<b>End Date:</b> " + endText + "<br><br>"
               + "We look forward to seeing you at the gym.<br><br>"
               + "Regards,<br>Gym Management Team"
    ]
}
```

3.3 Flow of the automation

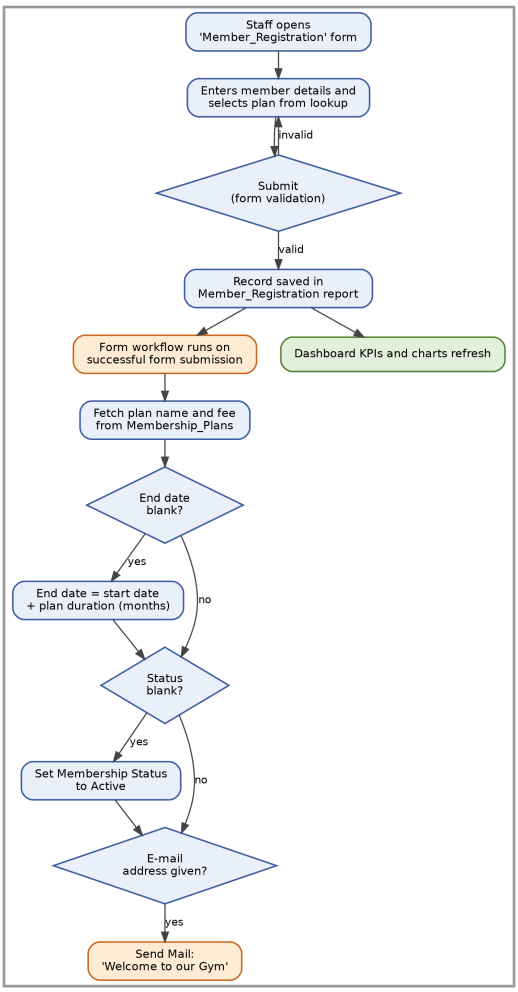


Figure 1: Flow of the member-registration workflow.

3.4 Testing and debugging output

The welcome e-mail is the proof that the script ran through all four steps. It shows the plan name and fee read from Membership_Plans (Platinum, Rs. 9500.00), proving that the lookup and fetch worked, and the start and end dates formatted as dd-MMM-yyyy, proving the date handling. The e-mail was received at 12:13 PM from the Zoho Creator mail service. The end date in this test (17-Mar-2027) was entered with the registration, so the script correctly left it unchanged.

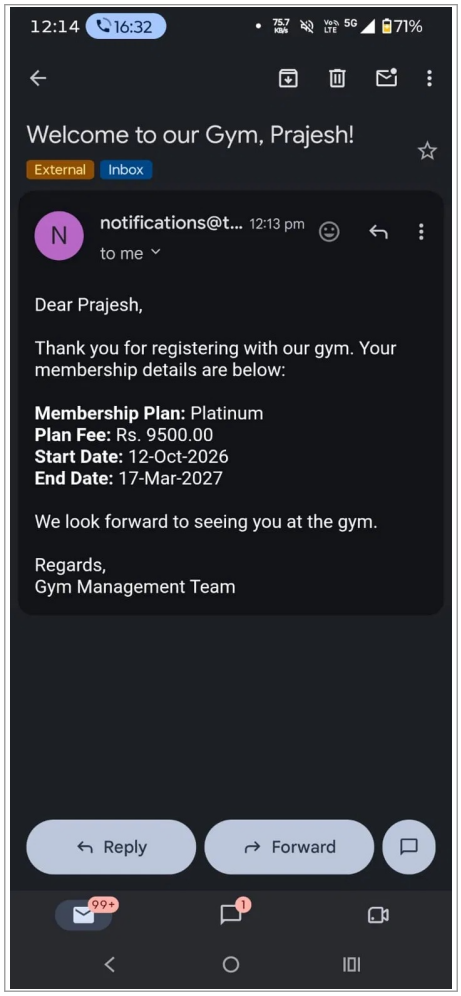


Figure 2: Welcome e-mail received after registration (subject "Welcome to our Gym", Platinum plan, Rs. 9500.00, 12-Oct-2026 to 17-Mar-2027).

Debugging approach. The Deluge editor checks the syntax before saving. Each step is guarded: the end date is calculated only when a start date and a valid plan exist, and the e-mail is sent only when an address exists, so the script does not fail on empty optional values.

4. Final Result

A form workflow with a Deluge script was built on Member_Registration. On every successful registration it calculates the end date, sets the default status and e-mails the member, with no manual work from the front desk. The delivered e-mail verified the workflow end to end.